Program : **Diploma in Textile Technology**	
Course Code : **3064**	Course Title: **Design and Structure of Fabrics**
Semester : **3**	Credits: **4**
Course Category: **Program core**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To impart the knowledge of textile designs used in the construction of fabric with their properties and uses.

Course Prerequisites

Topic	Course code	Course name	Semester
Basic knowledge on textile fibres		Textile fibres	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Construct Plain and its modifications	16	Applying
CO2	Construct twill designs and toweling fabrics	15	Applying
CO3	Illustrate various compound structured fabric designs	15	Understanding
CO4	Illustrate various types of crepe and perforated fabrics	12	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Construct Plain and its modifications**		
M1.01	Understand textile designs	3	Understanding
M1.02	Illustrate plain design and its modifications.	3	Applying.
M1.03	Understand Different types of drafts and its applications.	5	Applying.
M1.04	Understand lifting plan, tie-up, treadling plan and denting plan for various designs.	5	Applying.
Contents: Definition of design, draft, lifting plan and denting plan - Methods of representing design - Type of weave representation - warp overlap and weft overlap - Represent design, draft, lifting plan, Tie-up, treadling plan and denting plan on point paper - Types of draft - straight draft, broken draft, pointed draft, mixed draft - Plain design- its construction- Structural & decorative modifications of plain design - warp rib designs, weft rib designs and mat designs - rib and cord effects. Suitable drafts and denting plans			
CO2	**Construct twill designs and toweling fabrics**		
M2.01	Understand the construction and features of Twill design.	3	Understanding
M2.02	Construct the different twill designs with suitable drafts, lifting plan & tie up, treadling plan.	5	Applying.
M2.03	Understand the features of toweling fabrics	2	Understanding
M2.04	Construct the different types of toweling fabrics	5	Applying.
	Series Test -I	1	

Contents:

Twill designs: Construction - Features - Classification - Simple twill - Compound twill - Left hand twill - Right hand twill - Warp faced twill - Weft faced twill - Even faced twill - Steep twill - Flat twill. Modification of twill designs - Satin and sateen - Warp Cork screw and weft Cork screw designs - Combined twills - Figured twill - horizontal & vertical Zig zag twills - Diamond twills - Broken twills - Broken and reversed twill - Drafts, lifting plan & tie up, treadling plan for the above designs - Influence of yarn twist Direction on the prominence of twill lines.

Toweling fabrics: Features of Huck - a -

Back fabrics: Construction of Standard Huck - a - Back (10 X 10) - Devon's Huck - a - Back. Construction of ordinary honey comb - stitched honey comb - Brighton's honey comb designs - Drafts applicable to the above designs.

Features of pile fabric: Classification of pile designs - Construction of 3-pick terry, 4-pick terry and, 5-pick terry pile designs - Drafts applicable.

Corded velveteen designs: Differentiate warp pile and weft pile structures, velvet & velveteen.

CO3	**Illustrate various compound structured fabric designs**		
M3.01	Construct the Bedford cord design and draft applicable for each.	3	Understanding
M3.02	Understand the construction and features of various types of pique fabrics.	4	Understanding
M3.03	Understand the construction of fabrics by Double cloth principle.	4	Understanding
M3.04	Understand the importance and construction of extra warp and extra weft designs	4	Understanding

Contents:

General features and Construction of Bedford cord fabrics - plain faced - Twill faced - Wadded Bedford cord and develop their drafts - Functions of cutting ends, face ends and wadding ends

Features and Construction of various types of pique fabrics - Coarse cut pique, Fine cut pique - Wadded pique - Functions of cutting, face, and wadding end - System of Drafting and Denting applicable to above designs - Construction of Double cloth - Double width cloth - Tubular cloth -

Construct extra warp and extra weft designs - Importance of extra warp and extra weft figuring in ornamentation of fabrics.

CO4	**Illustrate various types of crepe and perforated fabrics**		
M4.01	Understand the Features and method of constructing crepe fabrics	3	Understanding
M4.02	Understand Features and Application of perforated fabrics	3	Understanding

M4.03	Understand the Principles of Cross weaving	3	Understanding
M4.04	Understand the construction of plain gauze and leno	3	Understanding
	Series test - II	1	
Contents Features of crepe fabrics - Different methods of constructing crepe fabrics and Designs - By adding marks to satin designs - Combining plain threads with a floating design - Inserting one design over another - Reversing a small repeat - Principle of seer sucker fabrics - Define perforated fabrics - Application of Perforated fabrics - Construct 3 X 3, 4 X 4, 5 X 5 Mock- Leno designs - Draft and Denting plan - Principles of Cross weaving - Application of Doup Healds - types of douphealds - Various types of sheds formed in cross weaving - Plain shed - Open shed - Cross shed - Construction of plain gauze & leno - thread diagram & graphical representation - Drafting and lifting plan for Plain Gauze & leno			

Text / Reference:

T/R	Book Title/Author
T1	WATSON - Textile Design and Color
R2	Gokarneshan N – Fabric Structure and Design.
R3	NISBERT - Grammar of Textile Design Mahajan Publishers, Ahmedabad

Online resources:

Sl.No	Website Link
1	http://freetexbooks.blogspot.in/2011/10/handbook-of-weaving.html
2	https://www.youtube.com/
3	http://textilelearner.blogspot.in
4	http://www.nptel.ac.in/courses